

CHANGHYUN OH

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EXPERIENCE

Soft Machines Lab (Advisor: Carmel Majidi) | Carnegie Mellon University

MS Researcher | August 2025 – Present

- Performed 2D Modeling and designing of soft inductive sensor elements using planar coils and magnetorheological elastomer.
- Performed testing, experimentation and characterization of the inductive sensor (INSTRON, robotic arm).
- Applied Machine Learning algorithm to effectively predict 2D location of the pressure point on the sensor.
- Manuscript in preparation - "(Machine-Learning-Based Continuous Contact Localization with an Inductive Tactile Sensor")

Soft Materials and Structures Lab (Advisor: Michael D. Bartlett) | Virginia Tech

Undergraduate Researcher | January 2024 – July 2025

- Optimized 3D printing settings and post curing processes for flexible resin and conducted standard dogbone modulus tests to obtain desired material properties.
- Designed, fabricated, and tested octopus-inspired switchable adhesives.
- Presented a poster at Adhesion Society Annual Meeting 2025 ("Octopus-Inspired Adhesives with Switchable Attachment to Challenging Underwater Surfaces").

1touch Project | Remote

Product Manager | July 2024 – Present

- Developing a mobile app to provide personalized experience with various soccer statistics and information.
- Creating new metrics that are interesting to users and relevant to evaluating players and teams.
- Determined project scope and designed the entire structure of the app.

Mechanical Engineering Capstone Design Project (Advisor: Michael D. Bartlett) | Virginia Tech

Active Team Member | August 2024 – May 2025

- Developed a continuous 90° peel test machine.
- Designed and constructed electrical system for powering and controlling speeds of the motors.
- Developed a Gannt chart and created a budget chart covering a whole academic year.

EDUCATION

B.S. in Mechanical Engineering – Robotics and Mechatronics, Virginia Tech

2019 – 2025

M.S. in Mechanical Engineering, Carnegie Mellon University

2025 – Present

SKILLS & TRAINING

- **Mechanical Design:** 3D CAD (SolidWorks), 3D printing (FDM, DLP, SLA), laser processing
- **Material Fabrication & Characterization:** Mold casting, Direct Ink Writing, soldering, INSTRON
- **Software:** Python, MATLAB, Arduino, STM32, KiCad, LTspice, fritzing

Honors & Awards

- James P Wightman Scholarship from Macromolecules Innovation Institute (MII) – July
- John G. Dillard Travel Award from MII – July 2024
- Dean's List with Distinction – 2024 Spring, 2024 Fall
- Dean's List – 2023 Fall

LEADERSHIP & OTHER EXPERIENCES

- Course Assistant (Mechanics I: 2D Design). (Fall 2026)
- Course Assistant (Mechatronics Applications in Mechanical Engineering). (Fall 2025)
- Outreach activity lead (Pathways for Future Engineers). (June 2025)
- Outreach activity lead (Black Engineering Excellence). (July 2024)